

Scott A. Hartman

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Teaching Experience

Instructional Faculty III, Department of Biology (formerly Integrative Biology), University of Wisconsin-Madison 2020-Present

Fall: Biology of the Dinosaurs (ZOO 275);
Comparative Anatomy of the Vertebrates (ZOO 430)
Spring: Biology in Popular Media (ZOO 110);
Comparative & Evolutionary Physiology (ZOO 611) and Lab (ZOO 612)
Summer, online: Introductory Ecology (ZOO 260);
Biology of the Dinosaurs (ZOO 275)

Graduate Lecturer for Department of Integrative Biology, University of Wisconsin-Madison 2016-2020

Fall: Comparative Vertebrate Anatomy (ZOO 430)
Spring: Comparative & Evolutionary Physiology (ZOO 611)

Graduate Teaching Assistant, Zoology and Geosciences, University of Wisconsin-Madison 2013-2018

Visualarium, Los Angeles, CA 2012
Advanced Creature Design – Anatomy for Artists

Education

PhD in Paleobiology, Department of Geoscience, University of Wisconsin-Madison August, 2020

Bachelors of Science, Zoology, University of Wyoming, June, 2013
Concentrations: Evolutionary Biology, Paleontology

Select Scientific Publications

(Google Scholar: <https://scholar.google.com/citations?user=C28Tm0QAAAAJ&hl=en>)

Grosmougin, M., Wang, X., Zheng, X... **Hartman**, S.A. and Xu, X., 2025. Forelimb feathering, soft tissues, and skeleton of the flying dromaeosaurid Microraptor. BMC Ecology and Evolution, 25(1), pp.1-25.

- Hartman**, S. A., Lovelace, D. M., Linzmeier, B. J., Mathewson, P. D., & Porter, W. P. 2022. Mechanistic thermal modeling of Late Triassic terrestrial amniotes predicts biogeographic distribution. *Diversity*, 14(11)
- Pittman, M., Kaye, T. G., Wang, X., Zheng, X., Dececchi, T. A., & **Hartman**, S. A. 2022. Preserved soft anatomy confirms shoulder-powered upstroke of early theropod flyers, reveals enhanced early pygostylian upstroke, and explains early sternum loss. *Proceedings of the National Academy of Sciences*, 119(47), e2205476119.
- Lovelace, D. M., **Hartman**, S. A., Mathewson, P. D., Linzmeier, B. J., & Porter, W. P. 2020. Modeling Dragons: Using linked mechanistic physiological and microclimate models to explore environmental, physiological, and morphological constraints on the early evolution of dinosaurs. *PLoS One*, 15(5), e0223872.
- Hartman**, S., Mortimer M., Wahl W., Lomax D., Lippincott J., Lovelace D. 2019. A new paravian dinosaur from the Late Jurassic of North America supports a late acquisition of avian flight. *PeerJ* 7:e7247
- Snively, E., O'Brien, H., Henderson, D.M., Mallison, H., Surring, L.A., Burns, M.E., Holtz Jr, T.R., Russell, A.P., Witmer, L.M., Currie, P.J. and **Hartman**, S.A., 2019. Lower rotational inertia and larger leg muscles indicate more rapid turns in tyrannosaurids than in other large theropods. *PeerJ*, 7, p.e6432.
- Wang, X., Pittman, M., Zheng, X., Kaye, T. G., Falk, A. R., **Hartman**, S. A., & Xu, X. 2017. Basal paravian functional anatomy illuminated by high-detail body outline. *Nature Communications*, 8, 14576.
- Hendrickx, C., **Hartman**, S.A. and Mateus, O., 2015. An overview of non-avian theropod discoveries and classification. *PalArch's Journal of Vertebrate Palaeontology*, 12(1).
- Ryan, M. J.; Russell, A. P., and **Hartman**, S. 2010. A New Chasmosaurine Ceratopsid from the Judith River Formation, Montana, In *New Perspectives on Horned Dinosaurs: The Royal Tyrrell Museum Ceratopsian Symposium* (Ryan, Chinnery-Allgeier, & Eberth eds), Indiana University Press, 656 pp
- Mayr, G., Pohl, B., **Hartman**, S., & Peters, S. 2007. The tenth skeletal specimen of *Archaeopteryx*, *Zoological Journal of the Linnean Society* V. 149, pp. 97-116
- Lovelace, D. M., **Hartman**, S. A. & Wahl, W. R. 2007. Morphology of a specimen of *Supersaurus* (Dinosauria, Sauropoda) from the Morrison Formation of Wyoming, and a re-evaluation of diplodocid phylogeny. *Arquivos do Museu Nacional*, Rio de Janeiro 65, 527-544

Select Scientific Presentations

- Hartman**, S. 2021. Thermal ecology drove survivorship patterns of terrestrial vertebrates across the End Triassic Extinction, Triassic Vertebrate Paleontology Meetup 2021.
- Whisenant, J., Kufner, A., **Hartman**, S. 2021. A Clarification of Rabbit Shoulder Musculature for Teaching Functional Anatomy, Experimental Biology Conference 2021, AAA.
- Hartman**, S. & Fitch, A. 2021. Operationalizing a Late Origin of Avian Flight, Northeast Natural History Conference.
- Hartman**, S., Mortimer, M., Lovelace, D. 2018. A testable macroevolutionary framework for character acquisition in the origin of avian flight, SVP 2018.

- Hartman, S., Mortimer, M., Wahl, W., Lomax, D., & Lovelace, D.** 2017. New information on a paravian theropod from the Morrison Formation, SVP 2017.
- Hartman, S.** 2016. Mechanistic physiological modelling predicts geographic distribution of Late Triassic tetrapods. In Annual Meeting of the Society of Vertebrate Paleontology, SVP 2016.
- Hartman, S., Lovelace, D., & Linzmeier, B. J.** 2015. Using ecological modelling to quantify thermal constraints on two Late Triassic dinosaurs. *Journal of Vertebrate Paleontology*. Vol. 35, online supplement.
- Hartman, S.** 2012. Investigating the impact of competing interpretations of pectoral girdle placement and appendicular function on sauropod head height. *Journal of Vertebrate Paleontology*. Vol. 32, Online supplement.
- Hartman, S.** 2007. Re-evaluating wing shape in *Archaeopteryx*: Information from the Thermopolis specimen. *Journal of Vertebrate Paleontology*, Vol. 27, Supplement to No. 3.

Select Museum Work

- Science Director, Wyoming Dinosaur Center, Thermopolis, WY 2005-2010
- Managed research program, collaborated with national and international institutions
 - Oversaw excavation of multiple dig sites in two states
 - Developed and expanded public outreach program
 - Responsible for specimen curation, specified & implemented new specimen database
 - Regular public presentations on diverse paleontological and general science topics
 - Designed, budgeted and executed an overhaul 15,000 square feet of displays
 - Mounted/remounted 17 skeletons for display
- Illustration and Design, University of Wyoming Geological Museum, Laramie, WY 2004-2005
- Crafted new signs to maximize current displays
 - Developed consistent design language to better convey information

Media Science Consulting

- BBC Studios 2023-Present
- Walking with Dinosaurs*
Primary paleontological consultant
- Netflix 2023
- The Dinosaurs*
Anatomy & character design
- BBC Natural History Division & Apple TV+ 2018-2023
- Prehistoric Planet*
Anatomy & character design

Cretaceous Films LLC	2010-2011
<i>Walking with Dinosaurs: The Movie</i>	
Anatomy, character design & locomotion	
Creative Differences	2009-2011
<i>Dinosaur Revolution</i>	
Anatomy, Locomotion, Paleobiology Fact Checking	

GRANTS & AWARDS (Total \$186,000)

UW-Madison Instructional Laboratory Modernization program	2024-2025
Awarded to modernize lab in Noland 315 & 321	
UW-Madison Instructional Laboratory Modernization program	2020-2021
Awarded to modernize ZOO 430/612 lab in Noland 321	
C.F. Schiesser Outstanding Student Research Paper Award, Geoscience, UW-Madison	2020
Awarded for Hartman, et al., 2019	
George J. Verville Award in Geology and Geophysics, UW-Madison	2018
Awarded for excellence as a graduate student in paleontology	
L&S Teaching Fellow, College of Letters & Science, UW-Madison	2017
Awarded for excellence in teaching	
Experiment.com funding for "Describing a Jurassic maniraptoran dinosaur from the Morrison Formation of North America"	2016
Communication-B Teaching Assistant Fellow, University of Wisconsin-Madison	2015
Awarded for sustained success in teaching undergraduates scientific writing in their majors	
Jurassic Foundation	2004
Awarded for CT scanning a new Jurassic coelurosaur	
B.O.C.E.S.	2001 & 2002
Development of human evolution display & reverse classroom human origins course	
Wyoming Geological Association	2000
Travel grant to Mexico City	